Worksheet 5.1 — Radian Measure

Grade 12 Advanced Functions (MHF4U) · Unit 5: Trigonometric Functions

The Big Ideas

A radian measures an angle by arc length; a full turn is 2π radians $= 360°$. Convert with $\frac{\pi}{180}$ or $\frac{180}{\pi}$, and use $s = r\theta$ for arc length.

- Degrees→radians: $\times\frac{\pi}{180}$. Radians→degrees: $\times\frac{180}{\pi}$.
- $30° = \frac{\pi}{6}$, $45° = \frac{\pi}{4}$, $60° = \frac{\pi}{3}$, $90° = \frac{\pi}{2}$.
- Arc length $s = r\theta$ (θ in radians).

Core Concepts

A natural angle unit

A **radian** measures an angle by arc length: one radian subtends an arc equal to the radius, and a full circle is 2π radians.

Converting

Since $180° = \pi$ radians, convert by multiplying by $\dfrac{\pi}{180°}$ or $\dfrac{180°}{\pi}$ as needed.

Arc and sector

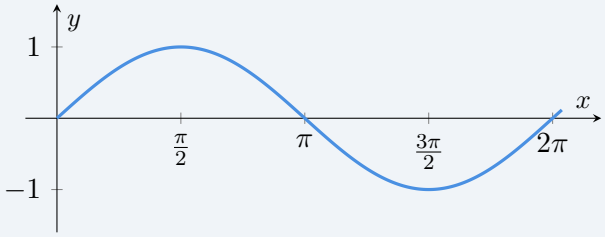
In radians the formulas simplify: arc length is $r\theta$ and sector area is $\frac{1}{2}r^2\theta$.

Worked Examples

Example 1 — Degrees to radians

Convert $180°$ to radians.

Solution. $180 \cdot \frac{\pi}{180} = \pi$. Radians are the natural axis for trig graphs:



Example 2 — Degrees to radians

Convert $90°$ to radians.

Solution. $90 \cdot \frac{\pi}{180} = \frac{\pi}{2}$.

Example 3 — Radians to degrees

Convert $\frac{5\pi}{4}$ to degrees.

Solution. $\frac{5\pi}{4} \cdot \frac{180}{\pi} = 225°$.

Example 4 — Degrees to radians

Convert $45°$ to radians.

Solution. $45 \cdot \frac{\pi}{180} = \frac{\pi}{4}$.

Example 5 — Arc length

Arc length for $r = 8,\ \theta = \frac{\pi}{4}$.

Solution. $s = r\theta = 8 \cdot \frac{\pi}{4} = 2\pi$.

Example 6 — Degrees to radians

Convert $15°$ to radians.

Solution. $15 \cdot \frac{\pi}{180} = \frac{\pi}{12}$.

Example 7 — Radians to degrees

Convert $\frac{\pi}{5}$ to degrees.

Solution. $\frac{\pi}{5} \cdot \frac{180}{\pi} = 36°$.

Example 8 — Degrees to radians

Convert $300°$ to radians.

Solution. $300 \cdot \frac{\pi}{180} = \frac{5\pi}{3}$.

Example 9 — Arc length

Arc length for $r = 12,\ \theta = \frac{\pi}{6}$.

Solution. $s = 12 \cdot \frac{\pi}{6} = 2\pi$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Convert $20°$ to radians.

Question 2

Convert $\frac{11\pi}{12}$ to degrees.

Question 3

Convert $80°$ to radians.

Question 4

Convert $\frac{7\pi}{9}$ to degrees.

Question 5

Arc length for $r = 9,\ \theta = \frac{2\pi}{3}$?

Question 6

Convert $40°$ to radians.

Question 7

Convert $\frac{5\pi}{9}$ to degrees.

Question 8

Convert $450°$ to radians.

Question 9

Arc length for $r = 20,\ \theta = \frac{\pi}{4}$?

Question 10

Convert $12°$ to radians.

Question 11

Convert $\frac{3\pi}{5}$ to degrees.

Question 12

Convert $105°$ to radians.

Question 13 — Challenge

A wheel of radius 0.5 m turns through 4π radians. How far does a point on the rim travel?

Question Recap & Answer Key

Question Recap

1. Convert $20°$ to radians.
2. Convert $\frac{11\pi}{12}$ to degrees.
3. Convert $80°$ to radians.
4. Convert $\frac{7\pi}{9}$ to degrees.
5. Arc length for $r = 9,\ \theta = \frac{2\pi}{3}$?
6. Convert $40°$ to radians.
7. Convert $\frac{5\pi}{9}$ to degrees.
8. Convert $450°$ to radians.
9. Arc length for $r = 20,\ \theta = \frac{\pi}{4}$?
10. Convert $12°$ to radians.
11. Convert $\frac{3\pi}{5}$ to degrees.
12. Convert $105°$ to radians.
13. A wheel of radius 0.5 m turns through 4π radians. How far does a point on the rim travel?

✓ Answer Key

1. $\frac{\pi}{9}$
2. $165°$
3. $\frac{4\pi}{9}$
4. $140°$
5. 6π
6. $\frac{2\pi}{9}$
7. $100°$
8. $\frac{5\pi}{2}$
9. 5π
10. $\frac{\pi}{15}$
11. $108°$
12. $\frac{7\pi}{12}$
13. $s = 0.5 \cdot 4\pi = 2\pi \approx 6.28$ m